Ultrasound Contrast
Gas Microbubbles in the Vasculature

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Abstract: Gas-filled microbubbles are currently in clinical use as blood pool contrast agents for ultrasound imaging. The goal of this review is to discuss the trends and issues related to these relatively unusual intravascular materials, which are not small molecules per se, not polymers, not even nanoparticles, but larger micrometer size structures, compressible, flexible, elastic, and deformable. The intent is to connect current research and initial studies from 2 to 3 decades ago, tied to gas exchange between the bubbles and surrounding biological medium, in the following areas of focus: (1) parameters of microbubble movement in relation to vasculature specifics; (2) gas uptake and loss from the bubbles in the vasculature; (3) adhesion of microbubbles to target receptors in the vasculature; and (4) microbubble interaction with the surrounding vessels and tissues during insonation.

Microbubbles are generally safe and require orders of magnitude lower material doses than x-ray and magnetic resonance imaging contrast agents. Application of microbubbles will soon extend beyond blood pool contrast and tissue perfusion imaging. Microbubbles can probe molecular and cellular biomarkers of disease by targeted contrast ultrasound imaging. This approach is now in clinical trials, for example, with the aim to detect and delineate tumor nodes in prostate, breast, and ovarian cancer. Imaging of inflammation, ischemia-reperfusion injury, and ischemic memory is also feasible. More importantly, intravascular microbubbles can be used for local deposition of focused ultrasound energy to enhance drug and gene delivery to cells and tissues, across endothelial barrier, especially blood-brain barrier.

Overall, microbubble behavior, stability and in vivo lifetime, bioeffects upon the action of ultrasound and resulting enhancement of drug and gene delivery, as well as targeted imaging are critically dependent on the events of gas exchange between the bubbles and surrounding media, as outlined in this review.

Key Words: microbubbles, ultrasound contrast, targeted imaging, focused ultrasound, drug delivery, gas exchange, perfluorocarbon

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Since the early insight of Gramiak and Shah,[1,2] who have discovered that "miniature bubbles" can serve as intravascular contrast agents for ultrasound imaging, studies of the behavior of gas-filled microbubbles in the vasculature have experienced steady growth. Initial interest was focused on blood pool ultrasound imaging,[3] supported by the suggestion that hydrodynamic behavior of gas-filled particles in the bloodstream is similar to the behavior of red blood cells (RBCs).[4] Gradually, blood pool microbubbles have transferred from early-stage preclinical testing to widespread clinical use, with millions of procedures performed each year in radiology, cardiology, and other medical fields. At the same time, there are 2 more areas where microbubbles in the vasculature are becoming increasingly attractive—one is enhanced energy deposition and drug delivery, and another one is targeted imaging. This review addresses some of the general scientific considerations for all of these scenarios, underlying particle behaviors, usefulness, applications, and limitations. The intent is to consider the existing and potential future trends. In this review, it is not possible to assess the biomedical microbubble field in its entirety, as it lists close to 10,000 references in PubMed. Thus many of the directions and particular studies have to be excluded. In the discussion of gas exchange phenomena, we minimize physical chemistry details and formula derivations; those details can be found within the cited articles.

MICROBUBBLES IN BLOOD VESSELS: HYDRODYNAMICS

Early studies, adapted from vascular physiology, have provided foundation for further research, development, and practical application of microbubbles in the vasculature. As the first step, optical microscopy of the fluorescent microbubbles in a rodent peripheral vasculature imaging model was used to monitor microbubbles that have been injected into the bloodstream, and their flow velocity and intravascular behavior were compared with that of the fluorescent RBCs.[4] The study was performed with air-filled albumin microbubbles (Albunex), which were the first to achieve US Food and Drug Administration approval as an ultrasound contrast agent medical device; it is expected that all other microbubble particles behave in a comparable manner. Overall, the mean values of linear velocity for microbubbles and RBCs in the vessels correlated tightly. Similar velocity distribution trend was observed within the arterioles both for bubbles and RBCs, with slower flow by the walls and somewhat faster velocity in the center of the vessel. It was the first demonstration that, even for larger vessels, such as arterioles and postcapillary venules, microbubbles may travel in proximity to the vessel wall, which implies their ability to interact with endothelium lining of the vessels. This information proved important for further applications of microbubbles, such as drug delivery, transfection, and molecular (targeted) imaging.

Next foundational study was performed with the second-generation ultrasound contrast Definity (perflutren) microbubbles.[5] These microbubbles lose gas slower than the air-filled version; they are based on perfluorocarbon gas, with much lower solubility in water and blood. Therefore, the lifetime in the bloodstream is much higher, and microbubble observation (again, by optical imaging in this study) can take place for longer periods. It was found that the centerline velocity of microbubbles and RBCs in the vessels was similar, likewise in venules and arterioles, confirming the usefulness of microbubbles as blood pool ultrasound

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imaging agents. More importantly for further research, it was found that, upon intra-arterial injection, microbubbles that were 5 μm in diameter or higher would not circulate freely; static particles were observed as lodged in the capillaries and small arterioles. Size distribution frequency of static microbubbles observed in spinotrapezius muscle by optical microscopy was highest at 6 μm particle diameter (almost 30% of total), with only 1% of the lodged bubbles under 5 μm in diameter.

To assure that ultrasound contrast agent is not lodged and keeps recirculating in the vasculature, particle size should be 4 μm or less, based on this limited rodent study data set. Alternatively, if microbubble lodging in the vasculature is desirable, for example, for opening up the endothelial lining barrier, larger particles might be preferred. Of course, not intra-arterial but intravenous route is used for contrast agent administration in clinical practice or preclinical research most often. Therefore, it was expected that, in the absence of intracardiac shunts, all of the intravenously injected larger particles would be trapped in pulmonary capillaries. However, this may not always be correct in humans, as it is now known that arteriovenous anastomoses may take place.[6] These may occur upon exercise, arterial hypoxemia, and may even take place in the apex of the lungs of patients in supine position. As the goal of perfusion imaging assumes unimpeded circulation of ultrasound contrast, one should reduce the microbubble size, preferably below 4 μm, and make sure that bubbles in vivo do not experience any significant growth. Large microbubbles might even get retained in the microvasculature as a tissue depot.[7] It is even more important lately, as US Food and Drug Administration has relaxed a prior restriction and removed a contraindication, allowing administration of microbubble contrast to patients who might have cardiac shunts. In any case, if microbubble size is properly controlled, bubbles circulate freely and do not get retained and immobilized even in the slowest disorganized neovasculature in the tumor, after intravenous bolus (see Supplement Video 1 http://links.lww.com/RLI/A578).

GAS EXCHANGE BETWEEN THE BUBBLE AND SURROUNDING BIOLOGICAL MEDIUM

The importance of gas exchange between the microbubbles and surrounding biological fluid (ie, blood) for contrast ultrasound imaging became clear very early after discovery of microbubble ultrasound contrast. It was found that simple air bubbles were rapidly losing gas and collapsing, making ultrasound contrast signal disappear. Loss of air from a bubble can be attributed to Laplace pressure,[8] where surface tension of a small shell-less gas bubble effectively squeezes air out to dissolve in the surrounding medium. As Laplace pressure is inversely proportional to the bubble radius (see Equation 1, where p is Laplace pressure, r is bubble radius, and σ is surface tension at gas-liquid interface).

$$p = 2\sigma/r \quad (1)$$

With small particle size and sizable surface tension at air-water interface, Laplace pressure is higher, and smaller bubbles disappear progressively faster; the rate of bubble disappearance is further accelerated as bubble radius is reduced, until complete disappearance, within seconds. Theoretically, a 10-μm microbubble, filled with air, in an air-saturated environment, will disappear within 7 seconds; if the bubble diameter is 1 μm, it will disappear in less than a second. It is even faster if gas concentration in the medium is lower than its saturation level.

To stabilize the bubble, a coating has to be applied to drastically reduce surface tension. A "regular" surfactant layer, for example, palmitic acid, was first of surfactant-stabilized clinical air-filled microbubble formulations, Levovist. It was approved in Europe and Japan.[9] This microbubble could at least make it across the lung vasculature without disappearance from the bloodstream, and thus became a clinical intravenous blood pool ultrasound contrast. Palmitic acid does not serve as a strong stabilizer; it only moderately reduces surface tension between air and water. Therefore, gas loss takes place even with these simple surfactant-coated bubbles. Such bubbles after preparation from a dry precursor are not stable on storage and have to be administered 2 minutes after preparation. When such loose-shell surfactant bubbles float to the top of an aqueous dispersion, they fuse with vial headspace air and disappear. Interestingly, albumin-shelled air bubbles, as discussed previously,[3] demonstrate excellent storage stability (years in sealed vials in aqueous albumin saline). This is because, unlike a thin and loose monomolecular layer of palmitic acid with poor intermolecular cohesion, albumin shell, constructed from denatured disulfide-crosslinked protein,[10] is thicker (approximately 15 nm thickness layer). One may suggest that "essentially solid" albumin shell should have mechanical properties similar to hard-boiled egg white. Because of hard shell presence, Laplace pressure is brought to zero, and bubbles can stay in the media not for seconds and minutes, but for up to several years, if the media is saturated with air and there is equilibrium among the gas in the bubble, the gas dissolved in the surrounding medium, and the headspace gas in the sealed vial.

Next issue with gas loss from the bubble has to do with the fact that even albumin shell is relatively thin and not a strong barrier for gas transfer. Therefore, when air-filled bubbles are administered in the bloodstream, they still collapse and lose gas quickly, especially considering rapid cycling of oxygen concentration in the surrounding medium as microbubbles move with blood through veins, lungs, large, and small arteries, and arterioles, capillaries, postcapillary venules, and back to veins. Loss of air is faster if an animal is on oxygen.[11] In that case, concentration of nitrogen dissolved in the bloodstream is low, and nitrogen from air in the bubble core is immediately extracted out, bubbles collapse, and become nonechogenic. Air-filled Albunex microbubbles, with human albumin shell, became the first intravenous clinical ultrasound contrast approved in United States in 1997, as a medical device for echocardiography. Because of its insufficient in vivo lifetime, as well as suboptimal ultrasound imaging presets available at that time, this agent did not perform well in the market and had to be replaced.

There are two options available to address this problem; their combination is also feasible. One approach is to use a thicker microbubble shell, made of a material with high cohesion between the shell molecules, ie, to use a biocompatible polymer, such as poly(cyanoacrylate), that is, surgical glue,[12,13] or poly(lactide-co-glycolide).[14] Several such attempts were made[15,16]; none of them made it to clinical practice so far. The thick and mechanically strong polymer shell does not easily collapse upon gas exchange; nondeformed shell supports retention of gas inside, even air or nitrogen. These particles are quite stable; weeks of storage in aqueous medium is possible; bubbles are stable during centrifugal washes in the presence of Triton X-100 detergent without drastic loss of gas.[17] To achieve efficient acoustic backscatter response from these microbubbles, acoustic pressure above threshold value may be required for ultrasound imaging; the microbubble shell has to "crack" so that the entrapped gas can compress and expand freely in response to the ultrasound field.[18]

The second approach, applied in microbubbles currently in widespread clinical imaging, with several millions of doses administered to patients every year worldwide, is the use of low-solubility fluorinated gases, such as SF_6 and perfluorocarbons. The approach is based on the fact that fluorinated gases have orders of magnitude lower solubility in water than air, so gas diffusion out of these particles is slower.[19] Thus, circulation time of up to several minutes after intravenous bolus is achieved. These agents are as follows: SonoVue/Lumason/BR1/sulfur hexafluoride; Definity/perflutren/MRX115; Sonazoid/NC100100/perflubutane; Imagent/AF0150/perflexan; and Optison/FS069/perflutren. The first 4 contrast materials are based on lipid shell design, and the last one, Optison, is based on albumin shell. A detailed review of the clinical microbubble formulations is provided, for example, by Frinking et al.[20] We need to note that fluorinated gases per se are not sufficient for extending the lifetime of microbubbles. Generally, surface tension of water

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at an interface with fluorinated gas may be only somewhat lower (not orders of magnitude, not even several-fold) than surface tension of air-water interface.[21] So, from Epstein and Plesset[8] and Equation 1, we must conclude that the stability of micrometer-size lipid and protein-shelled clinical microbubbles mentioned previously is also explained by the action of the shell, which reduces surface tension to zero. This implies that the solid spherical shell at the gas-water interface acts like an "arch" that resists external force. It essentially nullifies the squeezing action of air-water surface tension forces on the gas phase, bringing extra pressure on the gas bubble to zero, and gas can stay within the shell.

The use of fluorinated gases with lower solubility in water and in blood resulted in the widespread clinical success of these "second-generation" contrast microbubbles and current absence of air-filled bubble commercial formulations in the clinic. There are some other remarkable processes that take place upon administration of these particles into the bloodstream; they were discussed by Van Liew and Burkard,[22,23] and some even brought to clinical use by the efforts of Kabalnov et al.[24,25] It was proven, first theoretically and then confirmed experimentally, that dissolved air from the surrounding aqueous medium (including blood) would extract into the fluorocarbon gas-filled bubble, while fluorocarbon gas cannot exit the bubble fast enough, because of its poor solubility, to compensate. Extraction and diffusion of dissolved air from the aqueous medium into fluorinated gas bubbles can be explained from Gibbs-free energy (specifically, entropy) standpoint; air mixed into a fluorocarbon gas phase possesses lower overall energy level, which drives the direction of gas extraction process toward mixing. Surface tension and respective Laplace pressure (if present) may counter the entry of dissolved air; therefore, smaller bubbles, within approximately 1 to 2 μm threshold diameter, would not even expand much, or might still compress.

Larger bubbles may grow much bigger; as they expand, they may escape the action of Laplace pressure, as it is inversely proportional to bubble radius (see Equation 1).[26] Eventually, the extracted air, as well as the fluorinated gas, will diffuse out through the shell and dissolve in the surrounding medium. These bubbles will shrink, collapse, and disappear, but it may take much longer time overall. It will take much longer for the least soluble perfluorocarbon-based microbubbles; almost up to an order of magnitude diameter increase was computed in modeling studies, with many hours[27] and even days[28] of lifetime for perfluoropentane and perfluorohexane in air-saturated medium expected. The experimental testing in vivo did not confirm these drastically expanded ultrasound contrast lifetimes.[25] However, fluorinated gas-filled microbubbles at approximately 5 to 10 μm diameter or higher, once administered in the bloodstream, would very likely expand and grow within the vasculature. If they reach and exceed the diameter of blood capillaries, they might become transiently retained in the vasculature, until excess gas diffuses out, so the bubbles shrink and recirculate again, as was discussed previously.[5,7] There are 3 potentially important consequences to this situation.

1. The deposited microbubbles may block blood flow through the capillaries. This might or might not become a problem, depending on the administered dose and particular tissue, but it would be best to avoid. Certainly, deposited microbubbles would not be consistent with the definition of the blood pool freely circulating contrast material.
2. For targeted (molecular) imaging studies (see below, Targeting of Microbubbles to Vascular Biomarkers), nonspecific accumulation of microbubbles in the normal, nontarget tissues might become a problem for target/background ratio and limit the ability to detect and delineate the areas of targeted bubble accumulation, that is, areas of disease.
3. If microbubbles are lodged in the vessels as microplugs, in contact with endothelial lining on all sides, then significantly stronger bioeffects in response to ultrasound treatment may be expected, compared with the case where bubbles are transferred along the vessel with blood flow (see below, Bioeffects of Microbubbles: Focal Energy Deposition by Ultrasound). This may represent either a significant problem (in case of an imaging study, where any bioeffects should be avoided) or a potential advantage, if a goal of the intervention is to open up the endothelial lining of the vasculature so that a drug or a gene delivery system will be able to enter the tissue interstitial space in the ultrasound-treated area efficiently. The latter situation is most pronounced for the transient opening of blood-brain barrier (BBB).

A simple technique that can reduce expansion of bubbles because of extraction of dissolved air is to adjust the insoluble gas core composition, that is, to add air or nitrogen to the bubble composition before administration.[29] This approach had been patented more than 2 decades ago, with the general goal to minimize potential undesirable bioeffects of ultrasound contrast. Conveniently, all other fluorinated gas microbubbles manufactured upon real-life preparation technologies would have certain amount of another gas present, and thus were covered by the claims of that patent. The bubble formulation with the intentional use of this technology that was approved for clinical use first, Imagent (perflexan), was based on high vapor pressure of perfluorohexane (27 kPa at 25°C, boiling point at 56°C), and copresent with nitrogen (17:83 vol/vol) in gas phase of the vial. This material did not seem to have extended bloodstream circulation time in the clinic, and at current time, it is not distributed for clinical use. Currently, similar approach is successfully used in BR55 (VEGFR2-targeted)[30,31] and BR38 (blood pool contrast) microbubbles, where perfluorobutane gas (boiling point at −2°C) is comixed with nitrogen at approximately 35:65 vol/vol ratio.[32] To achieve tight control of gas phase composition within the vials of ultrasound contrast, repeated gas fill cycles were described.[33] Some of the clinically approved bubbles, such as SonoVue/Lumason and Sonazoid, are prepared by adding saline or water to the dry matrix with interspersed lipids. Headspace in the matrix vial is almost 100% fluorinated gas. However, it is highly likely that the supplied aqueous medium containers will have air as headspace, so the bubbles would be exposed to air-saturated aqueous medium from the point of their formation. Postformation filtration through "spike" filters that have the nominal 5 μm pore size will also take out larger particles that had absorbed air preferentially.

Sometimes, before intravenous infusion of microbubbles, they are diluted, 10-fold or more, with normal saline that will very likely contain air.[34] Therefore, air could be introduced into the bubbles before intravascular administration, and the chances of intravascular bubble growth within the vasculature would be lowered. In any case, it is important to recognize and address this as a potential source of reproducibility issues for contrast imaging as well as for endothelial barrier opening studies. It should be also taken into account whether the patient is breathing oxygen or if an experimental animal receives inhalation anesthesia based on oxygen gas,[30] because nitrogen copresence in these bubbles combined with blood that has lower levels of dissolved nitrogen will result in the reduction of bubble size and shift in acoustic response; the potential for therapeutic bioeffects will also be lower. A bubble that has a gas core mixture of air and perfluorocarbon may swell or shrink, depending on the concentration of dissolved air in relation to saturation level in the surrounding aqueous medium. The process can be observed by video microscopy in real time (Supplement Video 2 http://links.lww.com/RLI/A579). Gradual gas loss and bubble volume reduction leads to gradual contraction of the encasing lipid shell. Eventually, compressive stress forces acting on the shell overcome intermolecular cohesion of the shell molecules. At that point, parts of the shell "click" and form folded structures that extend out from the bubble surface into the surrounding aqueous medium. Brightfield microscopy can be used to monitor gas core, and fluorescence microscopy can image the shell, containing trace amount of fluorescent lipid (Fig. 1).

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MICROBUBBLE DESTRUCTION BY ULTRASOUND IMAGING

Destruction of microbubbles during ultrasound imaging takes place if acoustic parameters are above certain threshold values, depending on the acoustic pressure, frequency, and pulse repetition frequency. Generally, at higher ultrasound frequencies, for example, 7 MHz and higher, mechanical index (MI) values of 0.2 and lower are nondestructive to perfluorocarbon microbubbles in vivo.[35] Above the threshold, the increase of acoustic pressure respectively increases the rate of microbubble disappearance. Complete destruction of bubbles subjected to a single ultrasound pulse is not assured even at higher pressures, in case when the microbubble concentration is high. Optical microscopy observation of microbubbles in a model study where they were "glued" to a polystyrene dish solid support and subjected to diagnostic imaging ultrasound[36] has found that rapid destruction of microbubbles happens only if the bubbles are at a distance from each other, either because of the creation of the oversaturated "cloud" of dissolved air around these close-by particles and/or, more likely, because of the interaction of the secondary acoustic waves between the neighboring microbubbles, which changes their vibration pattern. At high bubble surface density (eg, 86%), when microbubbles were surrounded on all sides by other microbubbles and positioned at close contact or within micrometers of each other, multiple pulses at 2.5 MHz were necessary to initiate microbubble destruction at MI of 0.3, even with air-filled bubbles. Deflation of microbubbles was more pronounced at the edge of the bubble patch.[36]

Two distinct regimes of microbubble disappearance have been reported. At peak negative acoustic pressures between ~400–600 KPa the dissolution process was gradual. At 800 KPa bubbles were lost through rapid fragmentation, ie, inertial cavitation could take place.[37] This provides an ability to control microbubble behavior remotely, from outside of the body.

Gas core composition improves microbubble stability in response to insonation; if a poorly soluble perfluorinated gas is used instead of air, destruction process significantly slows down.[38,39] An intriguing result was observed with perfluorocarbon-filled microbubbles that were insonated in perfluorocarbon-saturated, air-depleted aqueous medium; in these insonation conditions and media, microbubble destruction did not take place at all. However, in the air-saturated aqueous medium microbubbles would be destroyed (Figs. 2, 3). Specifically, after an overnight equilibration of the perfluorocarbon microbubble-carrying dish in room air atmosphere, microbubbles started to behave as air-filled bubbles in air-filled medium and were destroyed within seconds in exactly the same insonation conditions (MI of 0.5, 2 MHz, 30 Hz).[40]

Destruction of microbubbles by ultrasound is actually an important imaging tool for tissue perfusion quantification. Initially, when ultrasound imaging of microbubbles required their destruction to detect them over the tissue background signal, intermittent destruction with variable interval between destructive pulses was used for myocardial perfusion assessment.[41] The idea was to destroy all of the bubbles present in the vasculature within the field of view by the ultrasound pulse, for example, at end-systolic time point, then perform repeated intermittent destructive imaging at the next, every other, every third, fourth, and fifth contraction, to obtain the relative influx rates of bubbles into every pixel of the imaged tissue and distinguish the area of perfusion defect. Currently, with the advent of low-power ultrasound contrast imaging techniques, which combine phase inversion and acoustic pressure modulation, there is no need for repeated destruction; an ultrasound pulse at high MI is followed by low-power imaging at MI as low as 0.06,[42] where microbubble entry into the field of view can be monitored in real time, at a dozen frames per second, or even faster. Tissue perfusion is assessed by video analysis, and perfusion defect information can be obtained from microbubble arrival time data, as well as from the refill signal intensity kinetic exponential fit analysis. Care should be taken to make sure low-power ultrasound is indeed nondestructive. A slight elevation of acoustic pressure may result in partial degradation of microbubbles and distortion of the replenishment signal curve, leading to incorrect assessment of perfusion. If the initial destructive pulse is insufficient to destroy all microbubbles in the field of view, replenishment curve will also be distorted; sometimes, it is recommended to extend the intensity and time of the destructive pulse to several seconds at permitted maximum to avoid such distortion and assure complete destruction of all microbubbles in the interrogated volume.

MICROBUBBLE CONTRAST GAS REMOVAL FROM THE BODY BY EXHALATION

After intravenous administration of microbubbles, fluorinated gas is known to leave the body relatively rapidly, via exhalation. For all commercially available fluorinated gas microbubble contrast agents, there are reports of gas chromatography studies of exhaled air; these reports are often included in the prescribing information file. For instance, 75% to 80% of SF_6 gas intravenously administered as SonoVue

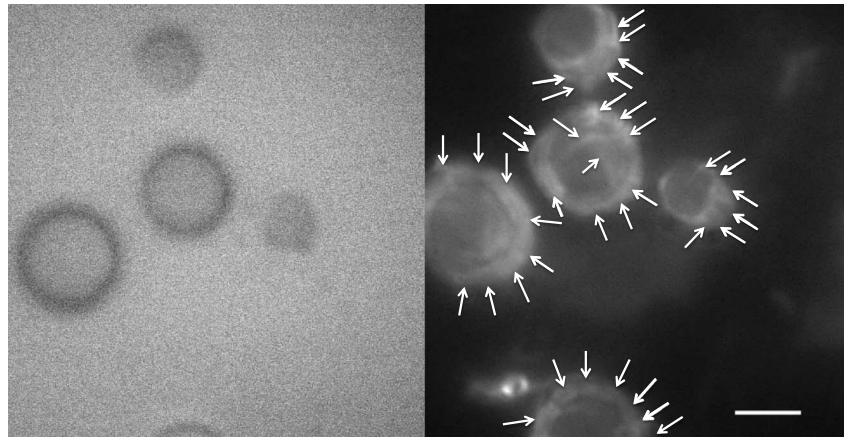


FIGURE 1. Brightfield (left) and fluorescence (right) microscopy of partially deflated C_{18}-lipid decafluorobutane/air microbubbles with excess surface area, forming folds that extend into surrounding aqueous medium. Bar, 10 μm. Arrows point at the lipid shell folds.

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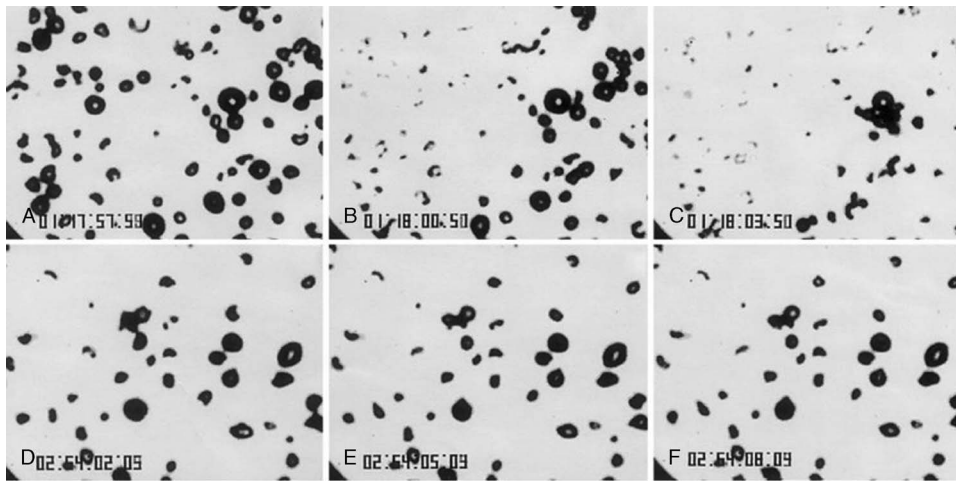


FIGURE 2. Video microscopic images of (A–C) Albunex and (D–F) Optison microbubble deflation in the ultrasound field. US system settings were 2 MHz, 30-Hz pulse repetition frequency, and MI of 0.5. Images were captured at 3-second intervals. Albunex microbubbles were in air-saturated saline; Optison microbubbles were in air-depleted perfluoropropane-saturated saline. Frame width was 110 μm. Copyright, The Association of University Radiologists, 2002. Reprinted with permission from Klibanov et al.[40]

microbubbles was exhaled and collected outside of the body within 11 minutes of intravenous bolus; almost half of the injected fluorinated gas was exhaled within 1 minute.[43,44] Gas loss and bubble collapse might take place in the bulk vasculature, where we can expect intense convection and rapid transfer of gas away from the thin diffusion layer around each microbubble, where gas solubility saturation is reached. At the same time, blood flow through the lungs is very high, as is their total surface area for gas exchange. Lung alveoli combination with microvasculature has evolved as a physiological tool with excellent efficiency of gas (oxygen and carbon dioxide) exchange. As fluorinated gas has some solubility in lipids, its ability to penetrate biomembranes is not zero; circulating bubbles lose fluorocarbon gas, and it leaves the body relatively quickly. It is yet to be understood if microbubbles that pass through the lung capillaries, or the dissolved gas in blood that had already left the bubbles, would be the ultimate source of fluorinated gas exhalation. It is already known that in the areas of slow and disorganized vasculature, such as tumor neovasculature, the retained (targeted) BR55 adherent microbubbles (lipid shell, nitrogen, and perfluorobutane core) are at times observed in the tumor for up to 30 minutes after intravenous bolus administration to patients with breast cancer, much longer than bulk circulation time.[45] It may be implied that microbubble gas exchange rate in slow-flow tumor vasculature is suppressed, as compared with large vessels with fast blood flow, as well as lungs.

A special case of a microbubble, NC100100/Sonazoid, has 2 distinct features. It has a rather poorly soluble perfluorobutane gas core, about an order of magnitude lower solubility than perfluoropropane. Sonazoid also has a solid monolayer shell of hydrogenated phosphatidylserine, which makes this agent an excellent target for phagocytic uptake in liver and spleen. In healthy volunteers, elimination half-life in exhaled air was reported as approximately 30 minutes; it was over 50 minutes in patients with impaired lung function.[46] This is a significantly slower elimination than for SF_6-loaded SonoVue microbubbles. One possible explanation is of course lower solubility of perfluorobutane as compared with SF_6. In addition to that, after phagocytosis, Sonazoid microbubbles are sequestered inside the macrophages and effective "shell thickness" of the bubble increases from 2 nm of the lipid monolayer to micrometers of the cell cytoplasm, in which the microbubble resides. Diffusion of poorly soluble gas through the cytoplasm and organelles of a macrophage will take much longer time; Sonazoid contrast was observed in patients' liver over an hour after intravenous bolus, most likely because of microbubble uptake into Kupffer cells.[47]

TARGETING OF MICROBUBBLES TO VASCULAR BIOMARKERS

A suggestion to target ultrasound contrast agents, as a general idea, not realized in practice at the time, was made in 1980s.[48] The formulation of microbubble particles with a thin elastic shell capable of acoustic backscatter response and encasing an insoluble gas necessary for sufficient in vivo lifetime was required; such particles appeared in 1990s. Next step was to attach targeting ligand molecules to the bubble shell, so that microbubbles would exhibit selective affinity to the

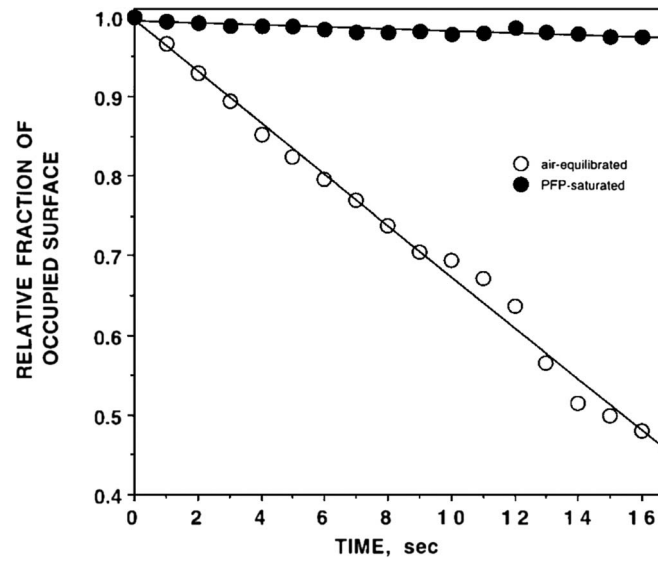


FIGURE 3. Kinetics of Optison microbubble deflation in the ultrasound field, presented as the relative fraction of the frame area occupied by microbubbles. Closed circles represent perfluoropropane-saturated saline medium; open circles represent air-equilibrated system. To achieve equilibration with air, the sample was left on the microscope stage in normal air atmosphere for 10 hours. The area occupied by microbubbles on the individual image frames was determined with the NIH Image software package (National Institutes of Health, Bethesda, MD) and normalized to the initial bubble surface coverage. Copyright, The Association of University Radiologists, 2002. Reprinted with permission from Klibanov et al.[40]

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disease biomarkers on the vessel wall luminal surface after intravenous administration,[49] or bind to the blood clot.[50] Initial preliminary tests with a model avidin-biotin pair had proven successful.[51,52] It was soon established that microbubbles can be observed by ultrasound imaging after they are attached to receptor-coated surfaces, with antigen-antibody model pairs,[53] antibody-mediated adhesion to activated (inflamed) endothelium,[54] and with peptide-receptor targeting,[50] selective adhesion of microbubbles to blood clots, inflamed endothelium, and tumor neovasculature. Below we discuss common features in these scenarios, and how gas exchange out of and into microbubbles may influence targeting to the molecular and cellular biomarkers of disease. The enabling factor in all of these studies is the ability of ultrasound imaging equipment to detect individual (single) ultrasound contrast microbubble particles in vivo, at significant depths (many centimeters), in real time, with excellent spatial resolution, noninvasively and nondestructively (see Supplement Video 1 http://links.lww.com/RLI/A578). As is clear from the contrast ultrasound video, observation of individual microbubbles in real time within the tumor vascular bed, nondestructive low MI imaging with complete background tissue suppression signal is achieved.[55] This imaging technique is fundamentally important for targeted imaging of microbubbles. One microbubble, with a micrometer radius, will have shell mass of a fraction of a picogram, and carry, for example, approximately 10^5 of antibody molecules. It will be able to adhere and remain on a target spot that has, for example, approximately 100 (or even less) biomarker receptor molecules per square micrometer,[56] and ultrasound imaging will be able to detect such bubbles in situ. Detection of individual adherent microbubbles by ultrasound imaging has been proven almost 2 decades ago.[57]

Next question is the force that will retain the adherent microbubble on the target surface. Initial experiments on the measurement of the force levels that can be obtained for ligand-decorated microbubbles were performed with the centrifuge force pull,[53] and then in a micropipette model system.[58] In the latter study, a ligand-microbubble and a receptor-coated bead were brought in contact with opposing micromanipulation pipets, and hydrostatic pulling pressure was then used to determine the forces necessary to pull the adherent bubble away from the bead; almost 100 nN force is required for the detachment of biotin-bubbles from avidin beads, and almost 20 nN for fluorescein-bubbles and antifluorescein antibody beads, which implies cooperative multipoint bonding between numerous ligand and receptor molecules for each particle. However, this study investigates axial pull detachment of a static adherent bubble, with the force applied normally to the contact plane. This is not a realistic model for the in vivo targeted adhesion and detachment application, where dynamic bubbles are carried in the tangential flow by the target vessel surface.

When a bubble is traveling through the vasculature, unless it is squeezing through the smallest capillaries, it will be intermixed with RBCs within the vessel lumen, carried by laminar flow in parallel to but mostly at a distance from the vessel wall, where the desired molecular biomarkers of disease are expressed, for example, on inflamed endothelium. Thus, efficacy of targeted adhesion of microbubbles in any vessels other than capillaries will be low, because most of the bubbles will travel in the center of the vessel and will not have an opportunity to touch the vessel wall target and interrogate it for presence or absence of the receptor biomarker of interest. This challenge can be addressed by the use of acoustic radiation force,[59,60] where the general area of interest is subjected to irradiation of low-amplitude ultrasound, which would not destroy microbubbles or make them lose gas, but force them from the center of the vessel to achieve direct proximity and ability to touch the vessel wall.

Considering the dynamics of microbubble particles in the blood flow during the act of targeted binding to biomarkers on the vessel wall, the following sequence of events is expected to take place. Particles move along the vessel, some in close proximity to the target vessel wall, and because of margination events,[61] they may touch the target surface. Assuming that the surface densities of ligand molecules on the bubble and receptor on the vessel wall are sufficiently high for ligand-receptor interaction, and contact time between these 2 molecules is in agreement with the bond formation kinetics, targeting ligand will bind to the biomarker receptor. It was estimated that the blood flow in a postcapillary venule microvessel generates shear force approximately 10^2 pN range on an adherent leukocyte[62] (cell diameter somewhat higher than for a microbubble). One might expect similar or lower shear to act on microbubbles in similar conditions in the vasculature.

As a microbubble is being translated along the flow direction, the newly formed ligand-receptor bond might either break or retain the bubble in place. The force necessary to break an individual ligand-receptor pair interaction at a molecular level, even in cases of strongest bonds known, is well under approximately 10^3 pN range, often close to 100 to 200 pN.[63] For the adherent bubble that projects into the flow of blood to resist the fluid shear force and retain it in place, a single ligand-receptor bond at moderate flow shear might be sufficient to hold the bubble in place, for the time necessary to form additional ligand-receptor pair bonds between the bubble and target surface. If viscous drag exceeds the ability of the ligand-receptor bonds to hold the bubble in place, firm adhesion and retention will not take place, and microbubble will be carried away with flow of blood, not permitting targeting.

For efficient targeting of microbubbles, we can take advantage of the tools that have evolved in nature, for example, for leukocyte adhesion to activated endothelium. Among these tools are the use of high contact surface area, proper shape, and the use of extended microvilli tethers. All these approaches are achievable with a simple trick, related to controlled partial gas loss by a microbubble before administration—to generate microbubbles that possess excessive shell surface area. Such microbubbles can be easily prepared from standard spherical microbubbles that are coated with a uniform lipid monolayer shell. Initial surface area of a sphere, $4\pi r^2$, becomes excessive, after gas core volume and radius reduction, and forms folds that project into the surrounding medium. For instance, assuming spherical shape of the gas core and initial uniform shell, a bubble that reduced from 2 μm radius to 1.6 μm (ie, shrunk by 20%) is expected to generate of approximately 18 μm^2 of excess lipid monolayer surface, nearly 50% of extra shell material. Controlled gas loss is accomplished by a transient pressurization, for 30 seconds at 1 atm.[64] Microbubble shell behavior after gas core loss can be observed in Figure 1 and Supplemental Video 2 http://links.lww.com/RLI/A579. Additional shell surface provides higher contact surface area between the bubble and the target, thus increasing the number of functional bond pairs between the bubble-bound ligand and the receptor target (Fig. 4). Cooperative multipoint interaction keeps the bubble immobilized on the target surface despite shear flow. In addition to that, microbubbles with extra shell surface can deform along the flow and assume elongated shape (Fig. 4). An elongated shape, even a teardrop, is a hallmark of adherent leukocytes that moderates viscous drag. Another reason to have micrometer-height features that project outward from the surface of targeted bubble is the force application arrangement; as the binding ligands are located at the "extender" arms, with length not much less than the bubble size, lever arm action will allow lesser force to hold the bubble in place (Fig. 5, left *vs* right). This extended tether approach is applied by leukocytes during their slow rolling in the vasculature.[65] Longer lever arm may help reduce the requirement on the ligand-receptor pair bond strength to bring it lower than 50 to 100 pN. In that case, a single bond might be able to retain the bubble in place despite shear flow for the time sufficient for more bonds to occur, so that the bubble would firmly attach to the target.

In large and small vessels on the arterial side, we expect to have much faster blood velocity and higher shear forces that will dislodge targeted microbubbles from the vessel wall. However, arterial blood flow is pulsatile; so at certain time points, for a fraction of a second, flow rate and shear will be much lower. If by that time targeted bubble will form multiple ligand-receptor bonds with the target site on the vessel

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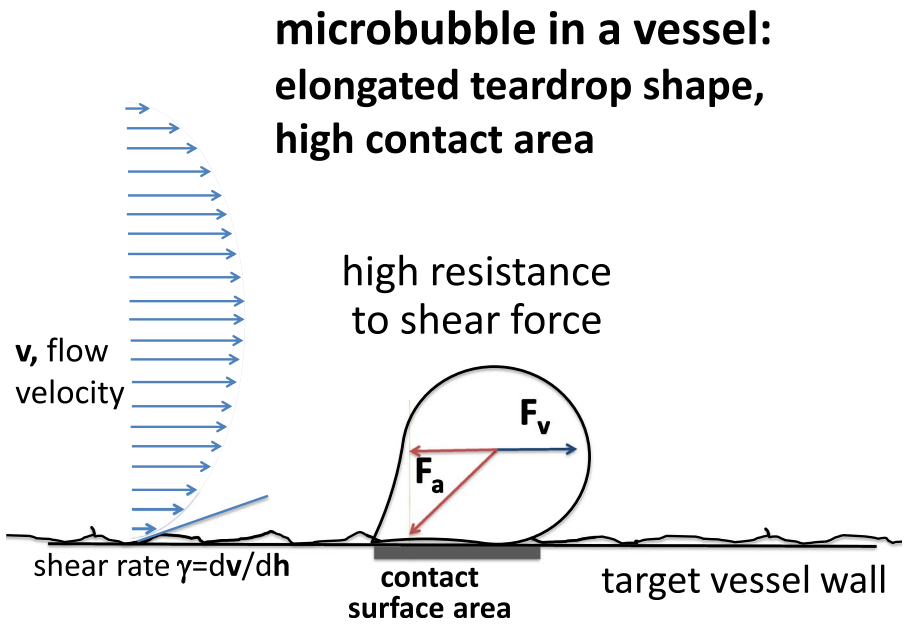


FIGURE 4. Targeted microbubble adherent to vessel wall biomarker receptors in shear flow. Microbubble assumes elongated teardrop shape that lowers viscous drag force F_v and increases contact surface area between the bubble and the target as well as the lever arm for the retaining ligand-receptor force F_a.

wall, it may be much more difficult to detach; it was found very early on that, if targeted attachment has taken place, for example, in static conditions, much higher shear flow will be required to detach these previously attached microbubbles.[53,56]

To summarize, modification of microbubble shell by partial gas loss is beneficial for targeted (molecular) imaging of vascular biomarkers; it provides beneficial biomechanics to improve targeted adhesion and retain microbubbles on the target surface that expresses molecular biomarkers of disease. We may note that microbubble gas loss may take place not just because of transient pressurization before administration in the bloodstream, as described previously. Excessive shell generation and membrane fold formation are likely to take place during microbubble circulation in the vasculature, as they gradually lose gas core.

BIOEFFECTS OF MICROBUBBLES: FOCAL ENERGY DEPOSITION BY ULTRASOUND

It has been noticed early that ultrasound demonstrates significant tissue and vasculature bioeffects when applied at high intensity in combination with intravascular microbubbles.[66] When microbubbles were destroyed in the vasculature, petechial hemorrhage in the insonated tissue was observed by intravital microscopy, if the acoustic pressure was sufficiently high. Microscopy has shown that microbubble/ultrasound

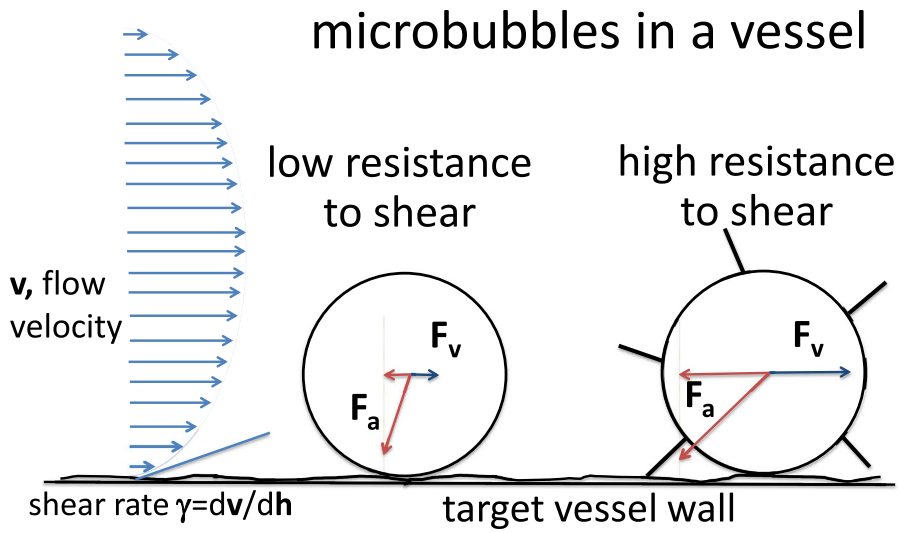


FIGURE 5. Targeted microbubble adherent to vessel wall biomarker receptors in shear flow. An adherent spherical microbubble (left) can only withstand small viscous drag force F_v because of small lever arm for the retaining ligand-receptor force F_a. In the presence of extended surface folds of the shell, the microbubble (right) can withstand much higher viscous drag force F_v because of longer lever arm for the retaining ligand-receptor force F_a.

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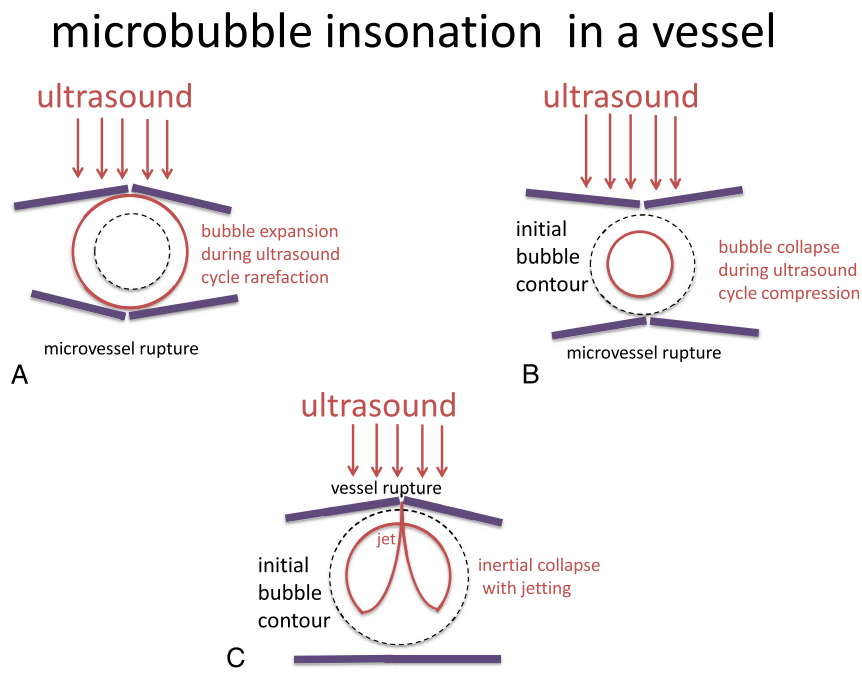


FIGURE 6. Events in the vasculature in response to microbubble insonation. A, Expansion of a microbubble during rarefaction portion of the ultrasound wave passage. B, Collapse of a microbubble during compression portion of the ultrasound wave passage. C, Collapse with jetting of a microbubble during compression portion of the ultrasound wave passage. Red line, microbubble during insonation. Black dashed line, before ultrasound action. Violet lines, vessel walls.

action can allow opening up of endothelial lining with extravasation of nanoparticles into the tissue interstitial space.[67] For diagnostic imaging, this is obviously an undesired bioeffect, so the strategy must be devised to avoid it. For therapeutic drug delivery, this might provide important practical benefit and open up numerous and important possibilities.

The mechanism of the observed phenomenon is relatively clear; as the microbubbles travel through the vasculature, they reach the capillaries. Then they are subjected to ultrasound action (Fig. 6). During the rarefaction phase (Fig. 6A), the expansion in a confined tube space forces the microbubble surface to forcefully press on the surrounding walls (ie, endothelial cells), deform the cells[68,69] and vessels,[70] and potentially break them. Likewise, the compression stage of the ultrasound wave cycle results in a rapid collapse and abrupt reduction of the bubble gas volume and diameter, leading to cell membrane defect formation (Fig. 6B), and sometimes, even asymmetric jetting sharp "cone" cavitational events (Fig. 6C) observed within the bubble.[71] Depending on the acoustic pressure applied, the velocity of the bubble wall movement during the collapse stage can reach 700 m/s.[72] Bubbles subjected to destructive insonation will reach capillaries only if the interval between the acoustic pulses is sufficiently long (it takes approximately 1 second or less for the blood flow to move from larger arterioles through small arterioles and into the capillaries). Thus, if higher frame rate (30 Hz) imaging is performed at high acoustic pressures, microbubble destruction will occur before the bubbles reach the capillaries, and ultrasound bioeffects will be minimized.[73] This information has provided an early confirmation of the importance of capillaries for microbubble-induced bioeffects of ultrasound. We may again (see Microbubbles in Blood Vessels: Hydrodynamics) consider the possibility that some of the microbubbles administered intravenously might bypass the capillaries in the lung and end up in the capillaries in other organs, subjected to ultrasound imaging. Large bubbles, especially filled with an undiluted insoluble fluorinated gas, will expand as they travel through the vasculature, by extracting dissolved air from the surrounding blood, and might reach the size that will stop them as soon as they enter the capillaries. Insonation of such particles that contact capillary wall surface on a perimeter is likely to lead to stronger bioeffects. Indeed, microbubbles with larger initial size have been reported to cause significantly stronger bioeffects than smaller bubbles.[74] Specifically, if the mean size of the microbubbles in the preparation was approximately 5.5 μm, with sizable amounts at 7 to 8 μm particles present within the size distribution, significant petechial hemorrhage was observed in the insonated kidney in a rodent model.[75] However, for the smaller microbubbles of the same Definity-style perfluoropropane-lipid microbubble formulation, with the mean diameter of the bubbles at 1.6 μm, and minimal amounts in excess of 5 μm, hemorrhaging was much lower at 3.6 MHz ultrasound frequency, and almost absent at 5.5 MHz. Interestingly, peak negative acoustic pressure values reported for both ultrasound probes used in this study were quite close, almost reaching −4 MPa (ie, inertial cavitation threshold exceeded by several-fold), and MI was 1.2 in both cases. With similar numbers of microbubbles administered, if we consider that the largest bubbles may be taken out by lung capillary filter, it is remarkable how small is the amount of larger particles that is sufficient to achieve these drastic bioeffects. Overall, controlling the size of microbubbles and assuring that larger particles are absent will be important both for imaging (to avoid undesirable enhancement of vascular permeability) and for drug delivery applications (to optimize the degree of opening of endothelial lining, and making it more reproducible). Addition of air or nitrogen to fluorinated gas microbubble core before administration may help suppress undesirable bubble growth (see Gas Exchange Between the Bubble and Surrounding Biological Medium). Uncontrolled growth of microbubbles in vivo should also be avoided, except in cases of tumor vasculature damage to the point of ablation, where they would be most desirable.[76–79]

The most exciting and potentially translatable phenomenon observed for microbubble insonation that could improve the treatment of tens of thousands of patients is the transient spatially targeted focal opening of BBB endothelial lining, with the goal of drug[80] and nucleic

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acid[81] delivery. In those scenarios, focused ultrasound treatment of microbubbles in the vasculature has to be performed in tightly controlled conditions to avoid inertial cavitation and accompanying destruction; those events might lead to microvessel damage sufficient for hemorrhaging, highly undesirable in the brain. To minimize unwanted bioeffects, low acoustic pressure ultrasound is applied. Acoustic characterization of the backscatter of activated bubbles (ie, the level of broadband acoustical emission monitoring) is used to avoid excessive ultrasound energy deposition.[82] This allows opening BBB for up to several hours, yet without harmful consequences. Shorter pulses used in rapid sequences can be applied if even shorter opening times are desirable (eg, if a drug to be delivered is present in the bloodstream for only several minutes).[83] Overall, "control knobs" for BBB opening include acoustic pressure, pulse length, pulse repetition frequency, and insonation frequency (although the latter may have to be limited to <0.4 MHz to focus through the human skull without significant energy loss). Microbubble dose and concentration in the vasculature will play a crucial role as well. Size distribution of bubbles may be of a significant importance to the levels of cavitation, as assessed in detail by Church and Miller.[84] In a clinical microbubble preparation, the fraction of larger bubbles (eg, 10 μm in diameter and above) is listed as relatively low, for example, under 5%, in a particle number distribution.[85] When volume distribution is considered, we may notice that the combined volume of bubbles exceeding 10 μm in diameter may be an order of magnitude higher than 5%. It was suggested already that the level of insonation bioeffects and BBB opening may be predicted from the volume of administered microbubbles.[86] We may also want to consider potential action of larger microbubbles, if they bypass the lung filter via shunts that might be present in the body. If they had all of their gas core material as perfluorocarbon before administration, these bubbles will grow and get transiently lodged in the tissues, which may result in some unpredictable insonation bioeffects; this might occur if shunts are present. Therefore, it is important to (1) premix air, nitrogen, or another gas with better solubility into the gas core, and/or (2) predilute microbubbles with air-saturated normal saline, as well as (3) generate microbubbles with narrow size distribution, devoid of particles in excess of 4 to 5 μm in diameter, by filtration, flotation, or other preparation means (see below, Microbubble Size Distribution Control, for microbubble size adjustment discussion). This way, the chances for uncontrolled bioeffects, which are already reasonably low, might be avoided completely. Microbubble agents such as BR38, with the core of nitrogen combined with decafluorobutane, might provide better predictability than currently approved materials that have a core of a single insoluble gas, allowing higher potential for growth. Alternative and simple technique would be to predilute single-gas bubbles with air-saturated normal saline before filtration and administration. At the same time, one has to consider that, if the patient is breathing oxygen, nitrogen gas will be rapidly extracted out through the lipid monolayer shell and bubbles will shrink. To assure procedure safety and reproducibility in the clinical setting, as well as the correct interpretation of experimental results in preclinical studies, the details of microbubble preparation and their gas core composition, as well as inhaled gas parameters and dissolved oxygen concentration in blood, has to be carefully controlled and reported.

In certain instances, it might not even be necessary to perform BBB opening to deliver payload of interest, for example, in case where endothelial lining of the vessel wall itself is the target. A recent study by Gorick et al[87] shows that a small dose of decafluorobutane microbubbles electrostatically coated with plasmid DNA, combined with very low acoustic pressure ultrasound, achieves efficient endothelium transfection without BBB opening. Thus, even the lowest insonation energy, which provides "soft massage" to endothelial lining by plasmid-carrying microbubbles, should be sufficient for safe therapeutic interventions, while maintaining endothelial tight junctions and BBB integrity.

MICROBUBBLE SIZE DISTRIBUTION CONTROL

As microbubble size is such an important parameter, tight control of size distribution is desirable. There are direct preparation approaches, for example, microfluidic chips that make microbubbles rapidly, one at a time, with essentially uniform size,[88] but those are yet to reach clinical practice. As clinical bubbles are not made that way, their size distribution is not narrow. To minimize the load of larger particles in some formulations, for example, SonoVue/Lumason, the microbubbles are filtered through a 5-μm mini-spike filter when pulled from the preparation vial into the syringe.[89] Although such filtration would remove "solid-shell" nondeformable particles, more elastic and flexible bubbles might squeeze through the filter pores, so this is not a universally applicable approach. The simpler technique that does not require additional pharmaceutical-grade approved sterile supplies, such as filters, and only takes moderate amount of time, yet allows complete removal of larger microbubbles from the preparation, is flotation.[35,90,91] Viscous drag force F_v for a solid spherical particle (ie, a microbubble with a solid shell) is proportional to media viscosity η and particle radius r (Fig. 7, Equation 2), as well as the particle velocity v.

$$F_v = 6\pi r\eta v \quad (2)$$

Particles in gravity field will move in accordance with their buoyancy in the medium, according to Archimedes. At the limiting (terminal) speed in the medium, viscous drag force equals buoyancy force, $F_b = F_v$, so there is no acceleration or deceleration, and flotation speed is constant. Density of microbubble core is approximately 2 to 3 orders of magnitude lower than density of water and can be neglected for simplicity; shell is thin and can also be neglected with reasonable precision. Buoyancy force F_b at normal gravity is proportional to the mass of water displaced by the bubble volume, that is, $F_b = 4/3\,\pi g r^3$, where g is standard gravitational acceleration. Resulting Navier-Stokes equation implies that largest particles travel upwards much faster than the smaller ones, with terminal velocity (limiting speed) proportional to square of the radius (Equation 3).

$$v = 2gr^2/9\eta \quad (3)$$

As viscosity of water (and aqueous buffers) is known, then rise time for microbubbles can be computed from terminal velocity, from general considerations. For instance, for viscosity of water, at normal gravity, in a cylindrical vial with the depth of liquid, for example, at 1 cm, 5 μm-diameter microbubbles will travel upwards at terminal velocity of approximately 0.8 mm/min. These bubbles, as well as all larger ones, will reach the top of that aqueous dispersion cylinder in approximately 12 minutes. At the same time, the microbubbles with 1 μm diameter will travel upwards at approximately 0.03 mm/min, so it will take approximately 300 minutes to travel 1 cm. Thus, only some of smaller bubbles will reach the top; this becomes a simple tool, where in a sealed vial, one can separate larger particles just by waiting for several minutes and collecting material from the bottom part of the vial. Instead of natural gravity, a centrifuge can be used,[92] so with proper equipment, instead of 10 to 15 minutes, you can complete size adjustment in a minute or less, depending on the bubble travel distance, centrifuge rotor radius, and rotation rate.

An important corollary to this propensity of microbubbles to float upwards is the need to agitate them for sampling and during transfer, for example, before and during an infusion from a syringe and via narrow tubing, especially if that tubing is horizontal. As flotation travel distance to the horizontal tube or a syringe barrel wall is short, larger bubbles will get to the tube top, and will stay there, if shear flow is insufficiently fast. It is not highly likely to happen in large vessels in vivo, because of the fast linear velocity of blood, and presence of hematocrit RBCs, that may provide additional shearing action, but it may take place during microbubble processing during preparation in large

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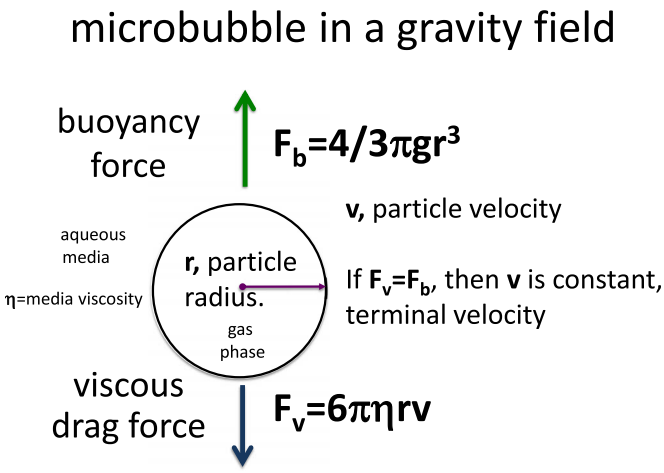


FIGURE 7. Microbubble upwards movement in the gravity field. A microbubble is subjected to buoyancy force F_b, essentially proportional to the volume of displaced water, which forces it upward. Movement of microbubble creates viscous drag, with viscous drag force F_v proportional to medium viscosity and particle radius; microbubbles accelerate until they reach limiting speed, then $F_b = F_v$.

batches, or if a horizontal tubing or syringe is used for intravenous infusion of microbubbles. Therefore, for the practical clinical considerations, to assure constant influx of microbubbles with unchanged size distribution during infusion, up to several milliliters of microbubble dispersion can be fed into the delivery line extension tubing connected to an intravascular catheter. Contrast agent is driven into the vasculature by saline infusion over the duration of the study, typically, a fraction of an hour. Available data imply that vertical (90 degrees) positioning of the tube would be preferred.[93] An alternative approach is to use a microbubble-loaded syringe in an infusion pump, as a more advanced technology that offers better reproducibility. A horizontally positioned syringe (or its barrel) is rotated to assure steady concentration and size distribution of administered microbubbles during infusion. This approach has been described in the literature in detail[94,95]; early patents on the subject must be expired by now.[96,97]

As was discussed previously, any microbubble handling (and in vivo exposure to biological fluids) implies gas exchange. Thus any dilution of microbubbles in aqueous media (including flotation for size adjustment purposes) that may expose bubbles to dissolved air, air-containing headspace, or another gas headspace will lead to gas extraction into or out of the bubbles, which may have an effect on bubble size distribution, strain on the shell, and changes in the particle stability profile during storage and in vivo.

CONCLUSIONS

Gas exchange between the bubbles and surrounding medium influences all aspects of microbubble behavior in vivo. We have demonstrated the fundamental importance of gas loss and intake for microbubble stability and behavior in vivo, stability in response to action of ultrasound, and retention of microbubbles in the small vessels. Bioeffects of insonation are also dependent on gas exchange, both via bubble size reduction rate as well as via retention in microvessel, which may lead to the desirable tissue drug and gene delivery. It may also cause undesirable and significant bioeffects, such as petechial hemorrhage (eg, with the administration of large bubbles filled exclusively with an insoluble gas). Even molecular ultrasound imaging with targeted microbubbles is influenced by gas exchange; partial gas loss improves the area of contact between the bubbles and the target. Extended structures (surface folds) on the microbubble shell create long lever arms, changing ligand-receptor force application biomechanics, and improving retention of targeted microbubbles on the surface that has molecular biomarker receptors. Meanwhile, gas loss from circulating bubbles via lungs, by exhalation, improves signal-to-noise ratio; rapid reduction of the background contrast level in the bloodstream allows targeted imaging within minutes of contrast administration. Gas exchange between microbubbles and surrounding media is rapid; it is dependent on many factors, so its precise control is complicated. However, lack of such control may lead to unexpected and not fully reproducible microbubble behavior. Direct measurement of gas composition of microbubbles within the bloodstream has not yet been reported and may be difficult to achieve, especially deep in the body, so we may not be able to control it easily. General concepts as presented previously provide some tools to manage microbubble behavior in the bloodstream toward the desirable trends, so that intravascular microbubbles will continue to advance to wider practical use in the field of medical imaging and drug delivery.

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